

**WASTE GENERATION ASPECTS:**

Waste stream assessment (WSA), waste generation and composition, waste characteristics (physical and chemical), health and environmental effects (public health and environmental), comparative assessment of waste generation and composition of developing and developed nations, a case study results from an Indian city, handouts on solid waste compositions. E-waste generation.

**MODULE – 2**

**WASTE GENERATION ASPECTS:**

**Waste Stream Assessment (WSA)**

Waste stream assessment (WSA) is a means to determine the basic aspects of quantity (i.e., the amount of waste generated in the community, both in terms of weight and volume), composition (i.e., the different components of waste stream) and sources of wastes. The information relating to these basic aspects of wastes is vital for making decisions about the SWM system, finance and regulations. Put differently, an assessment of waste stream is essential in the analysis of short and long-term problems within the local waste management system. It also helps in targeting waste management activities and setting goals for different elements of a waste management plan. Waste stream assessment, however, is not a one-time activity. It is a continuous and dynamic process, because the characteristics of wastes differ depending on the regions, communities, seasons, etc.

**Rationale for analysis**

The reasons for the analysis of waste composition, characteristics and quantity include the following:

- (i) It provides the basic data for the planning, designing and operation of the management systems.
- (ii) An ongoing analysis of the data helps detect changes in composition, characteristics and quantities of wastes, and the rates at which these changes take place, which facilitates effective implementation of management systems.
- (iii) It quantifies the amount and type of materials suitable for processing, recovery and recycling.
- (iv) It provides information that helps in deciding appropriate technologies and equipment.
- (v) The forecast trends assist designers and manufacturers in the production of collection vehicles and equipment suitable for future needs. In the absence of a reliable basic data, carrying out field investigations becomes necessary.

## Field investigation

Field investigations may take any one or a combination of the following forms:

(i) Waste sorting: Sorting of wastes into predetermined components takes place at disposal sites for weighing and sampling in order to determine the percentage of each component and the physical and chemical characteristics of wastes. It is carried out manually, and the sample size for analysis is between 100 and 150 kg. The implements required for the purpose include sorting table, measuring box, bins or boxes to contain sorted materials and platform weighing machine.

(ii) Vehicle weighing: Vehicles are weighed when they enter the disposal sites loaded, and exit the sites empty. The vehicle's front wheels are weighed first, followed by the rear wheels and the sum of the two gives the total weight. Weighing is carried out each day of the weighing period in order to determine the average weight. The weighing of loaded and unloaded vehicles is accomplished with a weighing scale or weighbridge. Ideally, the weighing scale should be operated during the entire period of operation of the disposal site, round the clock, if necessary. An electronic or a mechanical portable axle scale, with a capacity of 20 tonnes is suitable for the purpose. An electronic scale comprises two load-cell platforms and an electronic control and a display unit. The quantity of waste measured at disposal sites reflects a disposal factor rather than a generation factor, since the measurements do not include wastes that are:

- salvaged at the generation and disposal sites;
- disposed of in unauthorised places such as vacant plots, alleys, ditches, etc.;
- salvaged by collectors;
- lost during transport.

(iii) Field visits: This means visiting institutional and industrial sites to identify wastes being generated and disposal methods. Field visits involve visiting the facility, i.e., industry, institutions, etc., viewing the waste handling system and completing a questionnaire with the assistance of the plant manager or senior technical personnel who usually investigate wastes from industries and institutions. Collection of samples in sealed polythene bags follows for laboratory analysis to identify physical and chemical characteristics. Each sample may be in the range of 1.5 to 5 kg. An assessment of waste stream, in essence, helps us identify components that require improvement for effective implementation of waste management programmes (EPA, 1989 and 1995).

## WASTE GENERATION AND COMPOSITION

Information on waste quantity and composition is important in evaluating alternatives in terms of equipment, systems, plans, and management programs. For example, if wastes generated at a commercial facility consist of only paper products, the appropriate equipment is shredders and balers. Similarly, on the basis of the quantity generated, we can plan appropriate means for separation, collection, and recycling programs. That is to say, the success of SWM depends on the appropriate assessment of the quantity of waste generated.

### Waste generation

Waste generation encompasses those activities in which waste, be it solid or semi-solid material, no longer has sufficient economic value for its possessor to retain it. The processing of raw materials is the first stage when wastes are generated, and waste generation continues thereafter at every step in the process as raw materials are converted into final products for consumption. Figure 2.1 below shows a simplified material-flow diagram indicating the path of generation of solid wastes

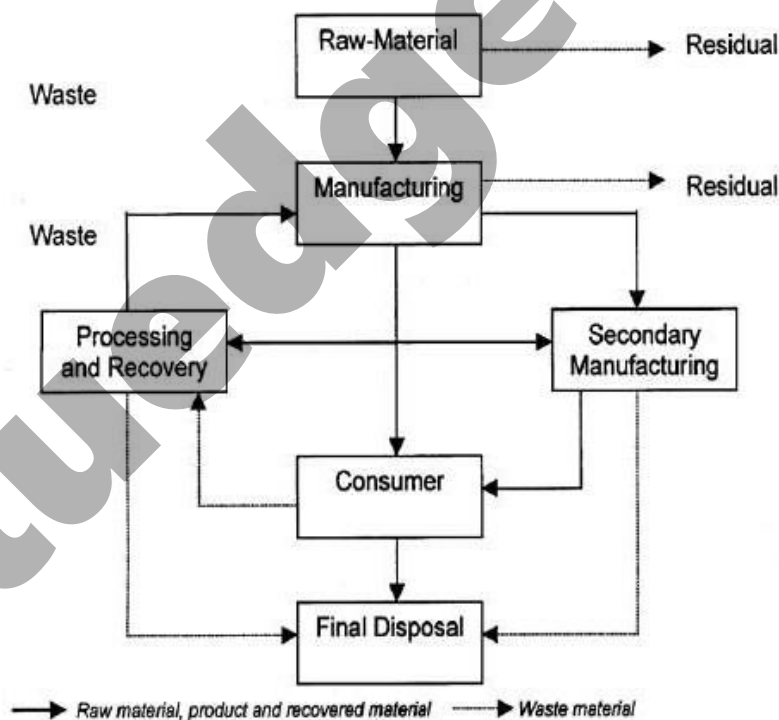


Figure 2.1 Material Flow and Waste

### Generation

Figure 2.1 suggests that we can reduce the amount of solid waste by limiting the consumption of raw materials and increasing the rate of recovery and reuse. There needs to be, therefore, a societal change in the perception of wastes. This sounds simple But, implementing changes in the society is

difficult, unless appropriate management solutions are provided. That said, we also must note that the changes in waste generation contribute to changes in waste composition.

### Waste composition:

Some of the general observations associated with the composition of wastes include the following: The major constituents are paper and decomposable organic materials. More often than not, metal, glass, ceramics, textile, dirt and wood form part of the composition, and their relative proportion depends on local factors. Average proportions of the constituents reaching the disposal sites are consistent and urban wastes are fairly constant although subject to long-term changes such as seasonal variations. Waste composition varies with the socio-economic status within a particular community, since income, for example, determines life style, composition pattern and cultural behaviour. Table 2.1 illustrates this phenomenon in India:

| Characteristics  | Low income                  | High income           | Comments                                                                                                                                    |
|------------------|-----------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                  |                             |                       | process.                                                                                                                                    |
| Moisture Content | 30 – 40%                    | 15 – 30%              | Moisture content depends largely on the nature of the waste, climate and collection frequency. Waste can dry out while awaiting collection. |
| Bulk Density     | 300 – 400 kg/m <sup>3</sup> | 150 kg/m <sup>3</sup> | Heavier waste may cost more to handle and difficult to burn.                                                                                |

**Table 2.1**  
**Typical Waste Composition: Low/High Income Population**

| Characteristics | Low income | High income | Comments                                                                                                 |
|-----------------|------------|-------------|----------------------------------------------------------------------------------------------------------|
| Paper           | 1 – 4%     | 20 – 50%    | Low paper content indicates low calorific value.                                                         |
| Plastics        | 1 – 6%     | 5 – 10%     | Plastic is low as compared to high-income areas though the use of plastic has increased in recent years. |
| Ash and Fines   | 17 – 62%   | 3 – 10%     | Ash and fines do not contribute to combustion                                                            |

Waste composition also depends on the moisture content, density and relative distribution of municipal wastes, as shown in Table 2.2 below, and is important for the characterisation of solid waste for most applications.

**Table 2.2**  
**Solid Wastes: Typical Composition, Moisture and Density**

| Components               | Mass % |         | Moisture content % |         | Density in kg/m |         |
|--------------------------|--------|---------|--------------------|---------|-----------------|---------|
|                          | Range  | Typical | Range              | Typical | Range           | Typical |
| Food wastes              | 6-26   | 14      | 50-80              | 70      | 120-480         | 290     |
| Paper                    | 15-45  | 34      | 4-10               | 6       | 30-130          | 85      |
| Cardboard                | 3-15   | 7       | 4-8                | 5       | 30-80           | 50      |
| Plastics                 | 2-8    | 5       | 1-4                | 2       | 30-130          | 65      |
| Textiles                 | 0-4    | 2       | 6-15               | 10      | 30-100          | 65      |
| Rubber                   | 0-1    | 0.5     | 1-4                | 2       | 90-200          | 130     |
| Leather                  | 0-2    | 0.5     | 8-12               | 10      | 90-260          | 160     |
| Garden Trimming          | 0-20   | 12      | 30-80              | 60      | 60-225          | 105     |
| Wood                     | 1-4    | 2       | 15-40              | 20      | 120-320         | 240     |
| Misc. Organic substances | 0-5    | 2       | 10-60              | 25      | 90-360          | 240     |
| Glass                    | 4-16   | 8       | 1-4                | 2       | 160-480         | 195     |
| Tin cans                 | 2-8    | 6       | 2-4                | 3       | 45-160          | 90      |
| Non-ferrous metals       | 0-1    | 1       | 2-4                | 2       | 60-240          | 160     |

| Components              | Mass % |         | Moisture content % |         | Density in kg/m |         |
|-------------------------|--------|---------|--------------------|---------|-----------------|---------|
|                         | Range  | Typical | Range              | Typical | Range           | Typical |
| Ferrous metals          | 1-4    | 2       | 2-6                | 3       | 120-1200        | 320     |
| Dirt, ash, bricks, etc. | 0-10   | 4       | 6-12               | 8       | 320-960         | 480     |

Note

that the density of waste changes as it moves from the source of generation to the point of ultimate disposal, and such factors as storage methods, salvaging activities, exposure to weather, handling methods and decomposition influence the density. In short, predicting changes of waste composition is as difficult as forecasting waste quantities.

### Factors causing variation

- As we know, wastes cause pollution. While the nature of wastes determines the type and intensity of pollution, it also helps us decide on the appropriate application, engineering design and technology for management. For example, the nature of wastes has implications for collection, transport and recycling. For effective SWM, therefore, we not only need information about the present but
- also the expected future quantity and composition of wastes. There are several factors, which affect the present as well as the future waste quantity and composition, and some of which are listed below:
- **Geographic location:** The influence of geographic location is related primarily to different climates that can influence both the amount of certain types of solid wastes generated and the collection operation. For instance, substantial variations in the amount of yard and garden wastes generated in Various parts of India are related to the climate. To illustrate, in the warmer southern areas, where the growing season is considerably longer compared to the northern areas, yard wastes are collected in considerably larger quantities and over a longer period of time.
- **Seasons:** Seasons of the year have implications for the quantities and composition of certain types of solid wastes. For example, the growing season of vegetables and fruits affect the quantities of food wastes.
- **Collection frequency:** A general observation is that in localities, where there are ultimate collection services, more wastes are *collected*. Note that this does not mean that more wastes are *generated*. For example, if a homeowner has access to only one or two containers per week, due to limited container capacity, he or she will store newspapers or other materials in some specified storage area. However, the same homeowner will tend to throw them away, if there is access to unlimited container services. In this latter situation, the quantity of waste generated may actually be the same but the quantity collected, as it relates to the frequency of collection, is considerably different.
- **Population diversity:** The characteristics of the population influence the quantity and composition of waste generated. The amount of waste generated is more in low-income areas compared to that in high-income areas. Similarly, the composition differs in terms of paper and other recyclables, which are typically more in high-income areas as against low income areas (see Table 2.1).

- **Extent of salvaging and recycling:** The existence of salvaging and recycling operation within a community definitely affects the quantity of wastes collected.
- **Public attitude:** Significant reduction in the quantity of solid waste is possible, if and when people are willing to change – on their own volition – their habits and lifestyles to conserve the natural resources and to reduce the economic burden associated with the management of solid wastes.
- **Legislation:** This refers to the existence of local and state regulations concerning the use and disposal of specific materials and is an important factor that influences the composition and generation of certain types of wastes. The Indian legislation dealing with packing and beverage container materials is an example. In short elements that relate to waste generation include land use characteristics, population in age distribution, legislation, socio economic conditions, household and approximate number.

## WASTE CHARACTERISTICS

In order to identify the exact characteristics of municipal wastes, it is necessary that we analyse them using physical and chemical parameters.

### Physical characteristics

Information and data on the physical characteristics of solid wastes are important for the selection and operation of equipment and for the analysis and design of disposal facilities. The required information and data include the following:

- **Density:** Density of waste, i.e., its mass per unit volume ( $\text{kg/m}^3$ ), is a critical factor in the design of a SWM system, e.g., the design of sanitary landfills, storage, types of collection and transport vehicles, etc. To explain, an efficient operation of a landfill demands compaction of wastes to optimum density. Any normal compaction equipment can achieve reduction in volume of wastes by 75%, which increases an initial density of  $100 \text{ kg/m}^3$  to  $400 \text{ kg/m}^3$ . In other words, a waste collection vehicle can haul four times the weight of waste in its compacted state than when it is uncompacted. A high initial density of waste precludes the achievement of a high compaction ratio and the compaction ratio achieved is no greater than 1.5:1. Significant changes in density occur spontaneously as the waste moves from source to disposal, due to scavenging, handling, wetting and drying by the weather, vibration in the collection vehicle and decomposition.

Note that:

- the effect of increasing the moisture content of the waste is detrimental in the sense that dry density decreases at higher moisture levels;

- soil-cover plays an important role in containing the waste;
- there is an upper limit to the density, and the conservative estimate of in-place density for waste in a sanitary landfill is about 600 kg/m<sup>3</sup>
- **Moisture content:** Moisture content is defined as the ratio of the weight of water (wet weight - dry weight) to the total weight of the wet waste. Moisture increases the weight of solid wastes, and thereby, the cost of collection and transport. In addition, moisture content is a critical determinant in the economic feasibility of waste treatment by incineration, because wet waste consumes energy for evaporation of water and in raising the temperature of water vapour. In the main, wastes should be insulated from rainfall or other extraneous water. We can calculate the moisture percentage, using the formula given below: A typical range of moisture content is 20 to 40%, representing the extremes of wastes in an arid climate and in the wet season of a region of high precipitation. However, values greater than 40% are not uncommon.

$$\text{Moisture content (\%)} = \frac{\text{Wet weight} - \text{Dry weight}}{\text{Wet weight}} \times 100$$

- **Size:** Measurement of size distribution of particles in waste stream is important because of its significance in the design of mechanical separators and shredders. Generally, the results of size distribution analysis are expressed in the manner used for soil particle analysis. That is to say, they are expressed as a plot of particle size (mm) against percentage, less than a given value.
- **Compressibility of MSW:** Degree of physical changes of the suspended solids or filter cake when subjected to pressure.

### Chemical characteristics

Knowledge of the classification of chemical compounds and their characteristics is essential for the proper understanding of the behaviour of waste, as it moves through the waste management system. The products of decomposition and heating values are two examples of chemical characteristics. If solid wastes are to be used as fuel, or are used for any other purpose, we must know their chemical characteristics, including the following:

- **Lipids:** This class of compounds includes fats, oils and grease, and the principal sources of lipids are garbage, cooking oils and fats. Lipids have high heating values, about 38,000 kJ/kg (kilojoules per kilogram), which makes waste with high lipid content suitable for energy recovery. Since lipids become liquid at temperatures slightly above ambient, they add to the liquid content during waste decomposition. Though they are biodegradable, the rate of biodegradation is relatively slow because lipids have a low solubility in water.

- **Carbohydrates:** These are found primarily in food and yard wastes, which encompass sugar and polymer of sugars (e.g., starch, cellulose, etc.) with general formula  $(CH_2O)_x$ . Carbohydrates are readily biodegraded to products such as carbon dioxide, water and methane. Decomposing carbohydrates attract flies and rats, and therefore, should not be left exposed for long duration.
- **Proteins:** These are compounds containing carbon, hydrogen, oxygen and nitrogen, and consist of an organic acid with a substituted amine group ( $NH_2$ ). They are mainly found in food and garden wastes. The partial decomposition of these compounds can result in the production of amines that have unpleasant odours.
- **Natural fibres:** These are found in paper products, food and yard wastes and include the natural compounds, cellulose and lignin, that are resistant to biodegradation. (Note that paper is almost 100% cellulose, cotton over 95% and wood products over 40%.) Because they are a highly combustible solid waste, having a high proportion of paper and wood products, they are suitable for incineration. Calorific values of oven-dried paper products are about half that for fuel oil, which is 44,200 kJ/kg.
- **Synthetic organic material (Plastics):** Accounting for 1 – 10%, plastics have become a significant component of solid waste in recent years. They are highly resistant to biodegradation and, therefore, are objectionable and of special concern in SWM. Hence the increasing attention being paid to the recycling of plastics to reduce the proportion of this waste component at disposal sites. Plastics have a high heating value, about 32,000 kJ/kg, which makes them very suitable for incineration. But, you must note that polyvinyl chloride (PVC), when burnt, produces dioxin and acid gas. The latter increases corrosion in the combustion system and is responsible for acid rain.
- **Non-combustibles:** This class includes glass, ceramics, metals, dust and ashes, and accounts for 12 – 25% of dry solids.
- **Heating value:** An evaluation of the potential of waste material for use as fuel for incineration requires a determination of its heating value, expressed as kilojoules per kilogram (kJ/kg). The heating value is determined experimentally using the *Bomb calorimeter test*, in which the heat generated, at a constant temperature of 25 C from the combustion of a dry sample is measured. Since the test temperature is below the boiling point of water (100 C), the combustion water remains in the liquid state. However, during combustion, the temperature of the combustion gases reaches above 100 C, and the resultant water is in the vapour form.

Table 2.3 shows the typical inert residue and heating values for the components of municipal solid waste (Tchobanoglous, et al., 1977):

**Table 2.3**  
**Typical Heating and Inert Residue Values**

| Component   | Inert Residue % |         | Heating Value (kJ/kg) |         |
|-------------|-----------------|---------|-----------------------|---------|
|             | Range           | Typical | Range                 | Typical |
| Food wastes | 2-8             | 5       | 3500-7000             | 4500    |
| Paper       | 4-8             | 6       | 11500-18500           | 16500   |
| Cardboard   | 3-6             | 5       | 14000-17500           | 16000   |
| Plastics    | 2-20            | 10      | 28000-37000           | 32500   |
| Textiles    | 2-4             | 2.5     | 15000-20000           | 17500   |

| Component               | Inert Residue % |         | Heating Value (kJ/kg) |         |
|-------------------------|-----------------|---------|-----------------------|---------|
|                         | Range           | Typical | Range                 | Typical |
| Rubber                  | 8-20            | 10      | 21000-28000           | 18500   |
| Leather                 | 8-20            | 10      | 15000-20000           | 17500   |
| Garden trimmings        | 2-6             | 4.5     | 2300-18500            | 6500    |
| Wood                    | 0.6-2           | 1.5     | 17500-20000           | 18500   |
| Glass                   | 96-99           | 98      | 120-240               | 140     |
| Tin cans                | 96-99           | 96      | -                     | -       |
| Nonferrous metals       | 90-99           | 96      | 240-1200              | 700     |
| Ferrous metals          | 94-99           | 98      | 240-1200              | 700     |
| Dirt, ash, bricks, etc. | 60-80           | 70      | 2300-11500            | 7000    |
| Municipal solid waste   |                 |         | 9500-13000            | 10500   |

Note that while evaluating incineration as a means of disposal or energy recovery, we need to consider the heating values of respective constituents (see Table 2.3).

For example:

- Organic material yields energy only when dry.
- The moisture content in the waste reduces the dry organic material per kilogram of waste and requires a significant amount of energy for drying.
- The ash content of the waste reduces the proportion of dry organic material per kilogram of waste and retains some heat when removed from the furnace.

**Ultimate analysis:** This refers to an analysis of waste to determine the proportion of carbon, hydrogen, oxygen, nitrogen and sulphur, and the analysis is done to make mass balance calculation for a chemical or thermal process. Besides, it is necessary to determine ash fraction because of its potentially harmful environmental effects, brought about by the presence of toxic metals such as cadmium, chromium, mercury, nickel, lead, tin and zinc. Note that other metals (e.g., iron, magnesium, etc.) may also be present but they are non-toxic. Table 2.4 shows the result of ultimate analysis of a typical municipal solid waste:

**Table 2.4**  
**Municipal Solid Waste: A Typical Ultimate Analysis**

| Element  | Range (%dry weight) |
|----------|---------------------|
| Carbon   | 25-30               |
| Hydrogen | 2.5-6.0             |
| Oxygen   | 15-30               |
| Nitrogen | 0.25-1.2            |
| Sulphur  | 0.02-0.12           |
| Ash      | 12-30               |

**Proximate analysis:** This is important in evaluating the combustion properties of wastes or a waste or refuse derived fuel. The fractions of interest are:

- moisture content, which adds weight to the waste without increasing its heating value, and the evaporation of water reduces the heat released from the fuel;
- ash, which adds weight without generating any heat during combustion;
- volatile matter, i.e., that portion of the waste that is converted to gases before and during combustion;
- fixed carbon, which represents the carbon remaining on the surface grates as charcoal. A waste or fuel with a high proportion of fixed carbon requires a longer retention time on the furnace grates to achieve complete combustion than a waste or fuel with a low proportion of fixed carbon.

**Table 2.5**  
**Municipal Solid Waste: A Typical Proximate Analysis**

| Components        | Value, percent |         |
|-------------------|----------------|---------|
|                   | Range          | Typical |
| Moisture          | 15-40          | 20      |
| Volatile matter   | 40-60          | 53      |
| Fixed carbon      | 5-12           | 7       |
| Glass, metal, ash | 15-30          | 20      |

To evaluate alternative processing and recovery options (e.g., incineration process), we need information on the chemical characteristics of wastes, and wastes can typically be a combination of Combustible and non-combustible materials.

Waste mismanagement, e.g., the practice of throwing wastes into streets, storm water drains, vacant land, etc., leads to breeding of disease vectors such as rats, with their attendant fleas carrying germs,

etc., which results in an epidemic such as plague, malaria, etc. This has an adverse impact on public health and the environment. We will look into the consequences of waste mismanagement on Public health and the environment.

## **HEALTH AND ENVIRONMENTAL EFFECTS**

An effective solid waste management system is necessary to avoid public health disasters, spread of disease by insects and vectors and adverse effect on water and air. Solid waste workers are the most exposed to the risks of parasitic infections and accidents, and therefore, a SWM system must include proper mechanisms to avoid these incidences. To the direct and indirect risks through accidents, exposure and spread of disease, we must add the effect of visual pollution caused by litter and nuisance created by smoke and dust at disposal sites.

### **Public health effect**

The volume of waste is increasing rapidly as a result of increasing population and improving economic conditions in various localities. This increased volume of wastes is posing serious problems due to insufficient workforce and other constraints in disposing of it properly. What are the consequences of improper management and handling of wastes? Consider the following:

**Disease vectors and pathways:** Wastes dumped indiscriminately provide the food and environment for thriving populations of vermin, which are the agents of various diseases. The pathways of pathogen transmission from wastes to humans are mostly indirect through insects – flies, mosquitoes and roaches and animals – rodents and pigs. Diseases become a public health problem when they are present in the human and animal population of surrounding communities, or if a carrier transmits the etiological agent from host to receptor.

**Flies:** Most common in this category is the housefly, which transmits typhoid, salmonellosis, gastroenteritis and dysentery. Flies have a flight range of about 10 km, and therefore, they are able to spread their influence over a relatively wide area. The four stages in their life-cycle are egg, larva, pupa and adult. Eggs are deposited in the warm, moist environment of decomposing food wastes. When they hatch, the larvae feed on the organic material, until certain maturity is reached, at which time they migrate from the waste to the soil of other dry loose material before being transformed into pupae. The pupae are inactive until the adult-fly emerges. The migration of larvae within 4 to 10 days provides the clue to an effective control measure, necessitating the removal of waste before migration of larvae. Consequently, in warm weather, municipal waste should be collected twice weekly for effective control. In addition, the quality of household and commercial storage containers is very significant. The guiding principle here is to restrict access to flies. Clearly, the use of suitable storage containers and general cleanliness at their location, as well as frequent collection of wastes, greatly reduces the population of flies. Control is also necessary at transfer stations, composting facilities and disposal sites to prevent

them from becoming breeding grounds for flies. Covering solid wastes with a layer of earth at landfill sites at the end of every day arrests the problem of fly breeding at the final stage.

**Mosquitoes:** They transmit diseases such as malaria, filaria and dengue fever. Since they breed in stagnant water, control measures should centre on the elimination of breeding places such as tins, cans, tyres, etc. Proper sanitary practices and general cleanliness in the community help eliminate the mosquito problems caused by the mismanagement of solid waste.

**Roaches:** These cause infection by physical contact and can transmit typhoid, cholera and amoebiasis. The problems of roaches are associated with the poor storage of solid waste.

**Rodents:** Rodents (rats) proliferate in uncontrolled deposits of solid wastes, which provide a source of food as well as shelter. They are responsible for the spread of diseases such as plague, murine typhus, leptospirosis, histoplasmosis, rat bite fever, dalmelonosis, trichinosis, etc. The fleas, which rats carry, also cause many diseases. This problem is associated not only with open dumping but also poor sanitation.

**Occupational hazards:** Workers handling wastes are at risk of accidents related to the nature of material and lack of safety precautions. The sharp edges of glass and metal and poorly constructed storage containers may inflict injuries to workers. It is, therefore, necessary for waste handlers to wear gloves, masks and be vaccinated. The infections associated with waste handling, include:

- skin and blood infections resulting from direct contact with waste and from infected wounds;
- eye and respiratory infections resulting from exposure to infected dust, especially during landfill operations;
- diseases that result from the bites of animals feeding on the waste;
- intestinal infections that are transmitted by flies feeding on the waste;
- chronic respiratory diseases, including cancers resulting from exposure to dust and hazardous compounds. In addition, the accidents associated with waste handling include:
- bone and muscle disorders resulting from the handling of heavy containers and the loading heights of vehicles;
- infecting wounds resulting from contact with sharp objects;
- reduced visibility, due to dust along the access routes, creates greater risk of accidents;
- poisoning and chemical burns resulting from contact with small amounts of hazardous chemical wastes mixed with general wastes such as pesticides, cleaning solutions and solvents in households and commercial establishments;
- burns and other injuries resulting from occupational accidents at waste disposal sites or from methane gas explosion at landfill sites;
- serious health hazards, particularly for children, due to careless dumping of lead-acid, nickel-cadmium and mercuric oxide batteries.

**Animals:** Apart from rodents, some animals (e.g., dogs, cats, pigs, etc.) also act as carriers of disease. For example, pigs are involved in the spread of diseases like trichinosis, cysticercosis and toxoplasmosis, which are transmitted through infected pork, eaten either in raw state or improperly cooked. Solid wastes, when fed to pigs, should be properly treated (cooked at 100 C for at least 50 minutes with suitable equipment).

**Environmental effect**

environmental effects such as the following:

(i) **Air pollution:** Burning of solid wastes in open dumps or in improperly designed incinerators emit pollutants (gaseous and particulate matters) to the atmosphere. Studies show that the environmental consequences of open burning are greater than incinerators, especially with respect to aldehydes and particulates. Emissions from an uncontrolled incinerator system include particulate matter, sulphur oxides, nitrogen oxides, hydrogen chloride, carbon monoxide, lead and mercury. Discharge of arsenic, cadmium and selenium is to be controlled, since they are toxic at relatively low exposure levels. Polychlorinated dibenzofurans (PCDFs), commonly called dioxins and furans, are of concern because of their toxicity, carcinogenicity and possible mutagenicity.

**Water and land pollution:** Water pollution results from dumping in open areas and storm water drains, and improper design, construction and/or operation of a sanitary landfill. Control of infiltration from rainfall and surface runoff is essential in order to minimise the production of leachate.

Pollution of groundwater can occur as a result of:

- the flow of groundwater through deposits of solid waste at landfill sites;
- percolation of rainfall or irrigation waters from solid wastes to the water table;
- diffusion and collection of gases generated by the decomposition of solid wastes.

The interaction between leachate contaminants and the soil depends on the characteristics of the soil. Soil bacteria stabilise **biochemical oxygen demand (BOD)**, i.e., the amount of oxygen required by micro-organisms to degrade organic matter, by anaerobic action, if toxic substances are in low concentration. The carbon dioxide produced keeps the pH level low, causing the water to dissolve minerals in the aquifers. Consequently, the change in groundwater quality may take place depending on the characteristics of the aquifer. Contamination can spread over considerable distances from the landfill, if the aquifers are of sand or gravel. In clayey soils, the rate of movement is greatly reduced. The capacity of clay to exchange ions restricts the movement of metal ions by capturing them in the soil matrix. Changes in its chemical characteristics are due to hardness, iron and manganese compounds.

**Visual pollution:** The aesthetic sensibility is offended by the unsightliness of piles of wastes on the roadside. The situation is made worse by the presence of scavengers rummaging in the waste. Waste carelessly and irresponsibly discarded in public thoroughfares, along roads and highways

and around communal bins (i.e., makeshift containers, without lids, used for the storage of residential, commercial and institutional wastes) gives easy access to animals scavenging for food. The solution to this social problem undoubtedly lies in the implementation of public education at all levels – primary, secondary, tertiary and adult, both short- and long-term, and in raising the status of public health workers and managers in solid waste management.

**Noise pollution:** Undesirable noise is a nuisance associated with operations at landfills, incinerators, transfer stations and sites used for recycling. This is due to the movement of vehicles, the operation of large machines and the diverse operations at an incinerator site. The impacts of noise pollution may be reduced by careful siting of SWM operations and by the use of noise barriers.

**Odour pollution:** Obnoxious odours due to the presence of decaying organic matter are characteristic of open dumps. They arise from anaerobic decomposition processes and their major constituents are particularly offensive. Proper landfill covering eliminates this nuisance.

**Explosion hazards:** Landfill gas, which is released during anaerobic decomposition processes, contains a high proportion of methane (35 – 73%). It can migrate through the soil over a considerable distance, leaving the buildings in the vicinity of sanitary landfill sites at risk, even after the closure of landfills. Several methods are available for control of landfill gas, such as venting, flaring and the use of impermeable barriers.

Evaluation methodology for generated solid waste involves analysis of landfill performance, the unit weight and compressibility, economic viability, MSW constituents, equations for evaluating MSW, data acquisition, source of evaluating that is study plan, demographic study, Questionnaire design to know who, what, where and why and statistics generation to analyse planning data.